

Futures First Energy Markets Internship — Final Report

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Over the internship I built an **Energy Dashboard** — a live energy-markets terminal (React front end, Express/Node API, offline Python analytics, deployed on Hugging Face Spaces across 10 sections and ~30 endpoints) — plus four quantitative studies around it. The aim was financial rather than technical: turn free public data — prices, EIA fundamentals, CFTC positioning, weather, news — into a read on crude oil and refined products that a desk could take seriously. Two rules run through everything: no model peeks at the future when it is tested, and every number links back to its source. The terminal is also deliberately upfront about what it *cannot* predict.

The work, and what the market taught me through it

The terminal. Ten live sections fed by Yahoo, Excel data from mentors, EIA, CFTC, weather and news sentiment. The design idea: a price alone tells you little — the same \$75 WTI means different things in a tight market and a glut — so every price is framed by the regime, the curve, inventories against their five-year norm, and the news flow.

The regime engine. Oil switches between states, and a signal that pays in one is noise in another. A rule-based classifier reads inventories, curve shape, volatility, trend and season; fair-value models on 13 spreads flag when one drifts too far from fundamentals, with a plain-English rationale for each call. The finding: reversion is real but regime-dependent — Brent–WTI returned toward fair value **83%** of the time across 108 signals (49% baseline); heating oil–gasoil 76%; the 3-2-1 refining margin 75%.

The historical backtest. The same idea over 5.4 years of daily WTI and Brent spread data, with a risk layer that cuts size or stands aside when the market turns violent. Fewer trades, better ones: **161 trades, profit factor 2.25, 67.7% win rate, ~\$320 per trade** before costs (a small bite at ~30 trades a year). Risk management isn't just protection — it is itself a source of return.

The intraday engine. The same idea on 15-minute crude spreads from a live company feed. Gross of costs it works; net of ~1bp slippage much of it disappears. An edge isn't an edge until it survives costs.

The inventory framework & Release Lab. The weekly EIA report is oil's biggest scheduled event, so it got the deepest treatment: a model of what the market should *expect* makes the **surprise** measurable, and a regime-weighted verdict says what should move. The Release Lab freezes a forecast before every Wednesday 10:30 ET release, then grades itself on a three-light scorecard after — forecast, surprise, and whether price actually followed. Across 162 material surprises, price followed the surprise **61.7%** of the time — real but modest, and it fades when OPEC, geopolitics or macro is running the market. The exception is Cushing near tank-tops — the April 2020 negative-price episode.

The HO–CL event study. Russia's invasion of Ukraine on the front-month diesel crack, from the 1-minute tape (with Hurricane Ida as a second, reusable case). The event didn't bump the crack, it *relocated* it: the day itself was a modest ~2.3 s.d. move, but the level shifted +33%, volatility at five times its prior pace, a peak near \$129/bbl — built over weeks, not in one violent tick.

The CFTC study. 547 weeks of Managed-Money positioning against WTI, tested every way I could. Positioning moves *with* price (+0.36 same week) but predicts nothing after it, and none of the 12 apparent edges survived a correction for testing many things at once. A crowding gauge, not a timing tool — and proving what *doesn't* work saves a desk capital.

What I take away

It all adds up to one way of working: **ask what kind of market this is, check inventories and the curve, compare the number to what was expected, judge how much it matters given everything else going on, then pick the trade — sized in case the market turns rough.** The beliefs behind it kept getting confirmed: markets react to surprises, not levels; context decides whether a signal is worth anything; the quieter numbers — time spreads, storage against normal — beat the headline price; and discipline beats cleverness, because testing only on unseen data is what separates a real edge from a lucky pattern. I also know where the work is still thin: the backtest is before costs, everything sits on a single five-year window, and my "expected" inventory number is my own model rather than the street's consensus. Those are what I would fix first.

What I enjoyed most was finding that every quantitative pattern has a concrete physical reason. Inventories absorb shocks because oil has to go somewhere — and the effect turns violent when Cushing's tanks near full. Time spreads beat the flat price because they ask the physical question: is a barrel today worth more than the same barrel next month? Brent–WTI has the most edge in my system because pipelines and shipping economics pull it back when it drifts. The 2022 diesel squeeze repriced over weeks because refineries and cargoes take weeks to rearrange. Positioning doesn't predict because a fund's position is a consequence of the move, not a cause of the next one. Even the plumbing follows the logic — gasoil quotes in \$/tonne because real cargoes trade in tonnes, and getting that wrong quietly corrupts everything downstream. Once I saw that chain from the physical world to the number on the screen, the models stopped feeling like statistics and started feeling like explanations.

Just as valuable were the habits. I learned real discipline — keeping to timings, being consistent, conducting myself well in the workspace — and waking up early for the office proved a welcome change of pace that streamlined the rest of my life around it. We didn't do actual trading, but I leave equipped with the fundamental market knowledge and the behaviour and discipline the job demands. Above all: the fundamentals matter as much as the finished product — a polished dashboard without sound reasoning underneath is a shell.